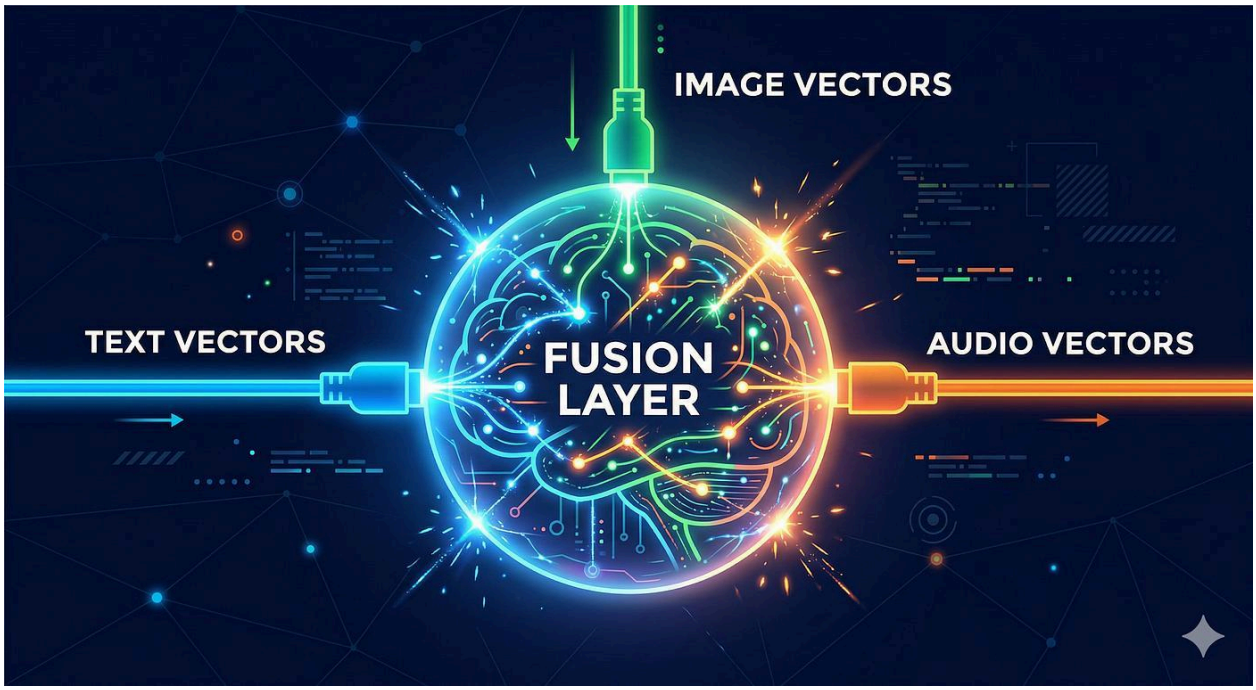
MULTI-SHIELD — Master Project Plan

Research blueprint · Problem definition · Architecture · Experiments · Impact · Publication strategy

1. The project in one sentence

MULTI-SHIELD investigates whether an AI system can make better decisions by identifying why its multimodal evidence is unreliable and choosing an appropriate response, rather than simply reducing the weight of unreliable information.



medium.com

Conceptual illustration of image, audio and text evidence entering a multimodal AI system.

Full title: MULTI-SHIELD: Failure-Aware Decision Policies for Reliable Multimodal AI under Compound Degradation

Domain: Artificial Intelligence and Machine Learning Research area: Robust Multimodal Learning, Trustworthy AI, Uncertainty Estimation and Selective Prediction.

The goal is not to build another multimodal classifier. It is to investigate whether a system that understands the type of evidence failure can choose a better intervention and make more reliable predictions.

2. What problem are we actually solving?

Modern AI systems can combine multiple sources of evidence:

- Images or video
- Audio or speech
- Text or contextual information
- In other applications, sensor measurements or structured records

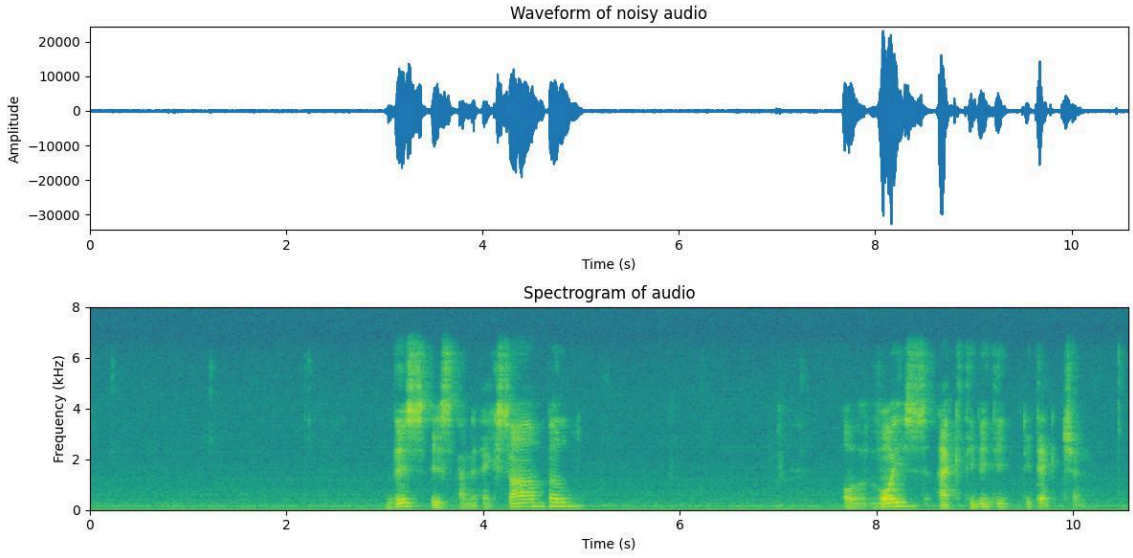
The problem is that these inputs are not always reliable.

Example: Three sources, three different problems



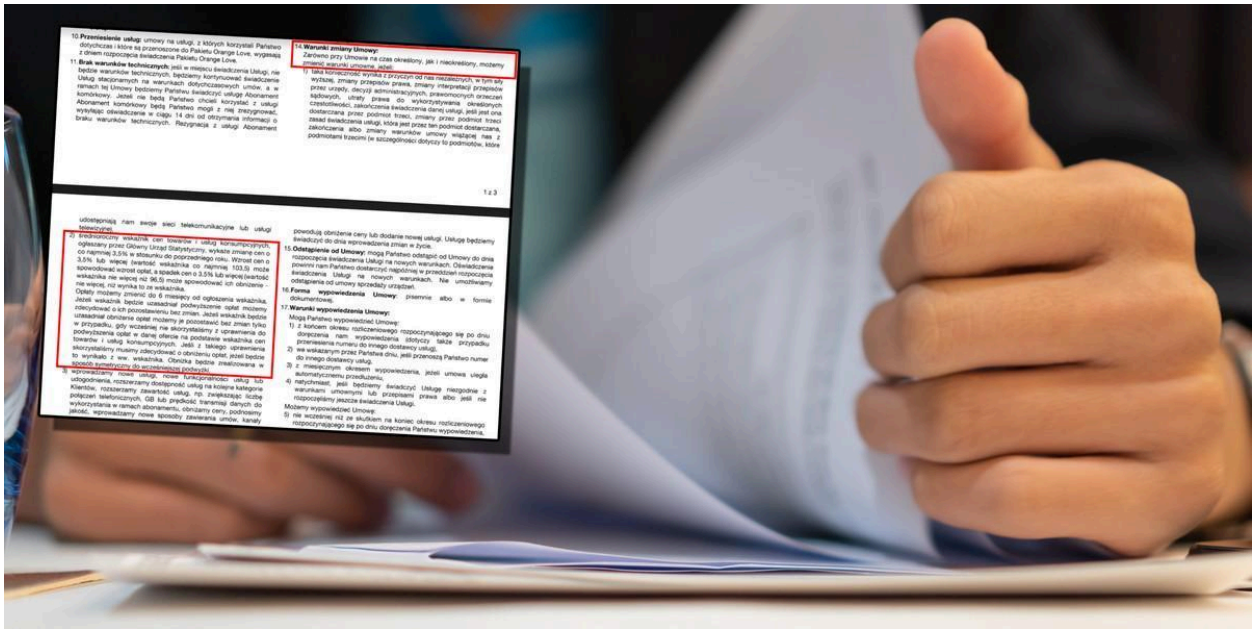
Image — corrupted

Part of the scene is blocked or distorted.



Audio — noisy or missing

Speech is difficult to understand, or no usable signal is available.



Text — contradictory

The text disagrees with the other evidence or the known context.

A conventional fusion model may combine these representations and produce a prediction anyway. That prediction might be correct, but the model may have no explicit mechanism to determine which intervention is appropriate for the particular failure.

This is the problem we want to investigate.

Why does it matter?

Research on multimodal fusion already documents performance degradation from noisy, incomplete and quality-varying inputs. A 2026 survey organizes these into overlapping reliability challenges and identifies unresolved questions in low-quality multimodal learning.

The practical issue is not merely that individual inputs can fail. The challenge is making a good decision when their quality varies, they interact, and the system cannot safely assume every input is trustworthy.

3. What causes this problem?

There are four major sources of failure.

A. Missing evidence

A camera feed drops, an audio recording is unavailable, or a data source fails to provide an input.

B. Degraded evidence

Noise, occlusion, blur, compression, interference or other damage makes an input less informative.

C. Conflicting evidence

Different modalities produce inconsistent interpretations, or one input contradicts information supplied by another source.

D. Compound degradation

Several problems occur together. For example, the image is corrupted, the audio is missing, and the text is contradictory.

The underlying cause is a mismatch between the assumptions of the system and the quality of the evidence it receives. The model may be trained on mostly usable inputs but encounter a combination of failures during deployment.

A critical complication is that disagreement does not necessarily mean corruption. A person can smile while expressing sadness. The system must not automatically suppress a modality simply because it differs from the others.

This distinction will be an important experimental question in MULTI-SHIELD.

4. What are the existing approaches?

We are not starting from zero. Several established approaches address parts of this problem.

Approach	How it works	Limitation we will investigate
Fixed fusion	Combines all modalities using a fixed architecture	May not adapt adequately to sample-specific failures
Adaptive weighting	Changes each modality's contribution according to estimated reliability	Does continuous weighting suffice for severe or interacting failures?
Missing-modality learning	Trains the model to handle absent inputs	Does not automatically solve noise, semantic conflict and corruption together
Uncertainty-awar e fusion	Uses uncertainty estimates to influence predictions or fusion	Uncertainty alone may not identify which intervention is appropriate
Restoration / imputation	Reconstructs missing or damaged information	Reconstructed features may be inaccurate or misleading
Selective prediction	Abstains when prediction risk is too high	Must be integrated with the reliability-management decision

These are not strawman alternatives. For example, DMS estimates confidence, uncertainty and semantic consistency to dynamically weight modalities. PRIME explicitly diagnoses modality reliability, restores representations, reassesses them and then fuses them. MCCF addresses uncertainty calibration under missing modalities.

What does this mean for our project?

We cannot claim novelty for simply:

- Estimating reliability
- Assigning dynamic weights
- Restoring representations
- Detecting conflict
- Adding an abstention mechanism

Those capabilities already have relevant prior art.

Our proposed distinction is to investigate whether an explicit failure-state representation can select among different interventions more effectively than a strong continuous-weighting approach, especially when several failures interact.

That is a hypothesis, not yet a proven research gap.

5. The core of MULTI-SHIELD

The core is a three-part decision process:

1. Diagnose

What is the condition of each input? Is it missing, noisy, corrupted, contradictory or apparently healthy?

2. Decide

Which intervention is most useful given the estimated failure state, the task and the cost of making a wrong decision?

3. Act and evaluate

Apply the intervention, make a prediction or defer, and measure whether the decision improved.

The four candidate actions

REWEIGHT — use less, not none

Reduce a modality's contribution when it remains informative but is less reliable than the alternatives.

SUPPRESS — exclude unreliable evidence

Mask a modality when allowing it into the decision is likely to be harmful.

REPAIR — attempt recovery

Reconstruct a degraded representation using information from other modalities, then reassess its reliability.

DEFER — decline an unsafe prediction

Abstain or request additional evidence when the remaining information does not justify a reliable decision.

These actions are candidates, not predetermined answers. The correct response depends on the context. A missing modality cannot be repaired in the same way as a slightly noisy one, and a repaired representation should never be assumed reliable merely because reconstruction succeeded.

6. How the architecture works

Here is the proposed end-to-end architecture.

Input layer

Image · Audio · Text

Modality encoders

Convert each input into a numerical representation

Evidence assessment

Confidence · Uncertainty · Signal quality · Cross-modal relationships

Failure-state representation

Estimate the likely condition of each modality and the joint evidence

Policy router

Select an intervention based on estimated failure state and expected utility

Reweight

Suppress

Repair

Defer

Decision and audit output

Prediction or abstention · Confidence · Action taken · Reason · Latency

What happens inside each layer?

Layer 1 — Encoders: Pretrained or lightweight encoders convert image, audio and text into numerical features. We do not need to train a giant foundation model from scratch.

Layer 2 — Evidence assessment: We calculate candidate diagnostic signals such as predictive entropy, epistemic uncertainty, feature-quality indicators and cross-modal agreement. We must test whether each signal adds useful information.

Layer 3 — Failure-state estimation: A classifier or probabilistic model estimates the likely condition of each modality. The output should include uncertainty about the diagnosis itself.

Layer 4 — Policy router: A learned policy or carefully controlled policy-selection mechanism selects an action. The policy must be trained and evaluated fairly, not given perfect failure labels at inference time.

Layer 5 — Decision and audit: The model makes a prediction or defers. The dashboard shows what happened, which action was selected, and how the result compares with baseline systems.

One essential rule: a disagreement score alone cannot distinguish genuine semantic divergence from corruption. That distinction requires contextual information, appropriate labels and experiments.

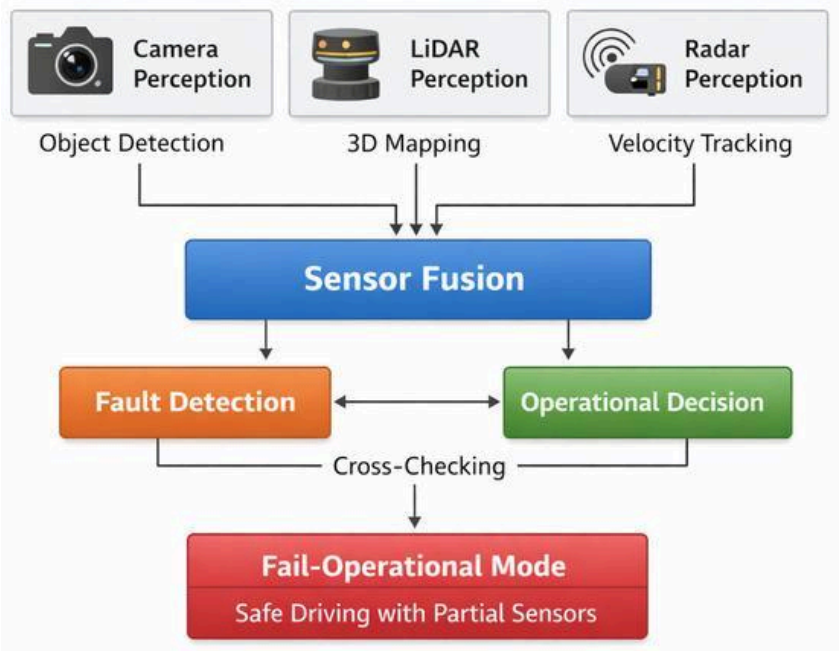
7. Who will use MULTI-SHIELD?

MULTI-SHIELD is a research framework, not yet a standalone product. Its eventual users depend on the application built around it.



1. Assistive-technology developers

Build systems that combine camera, speech and text inputs while detecting when evidence is inadequate. This is a strong demonstration application for inclusive technology.



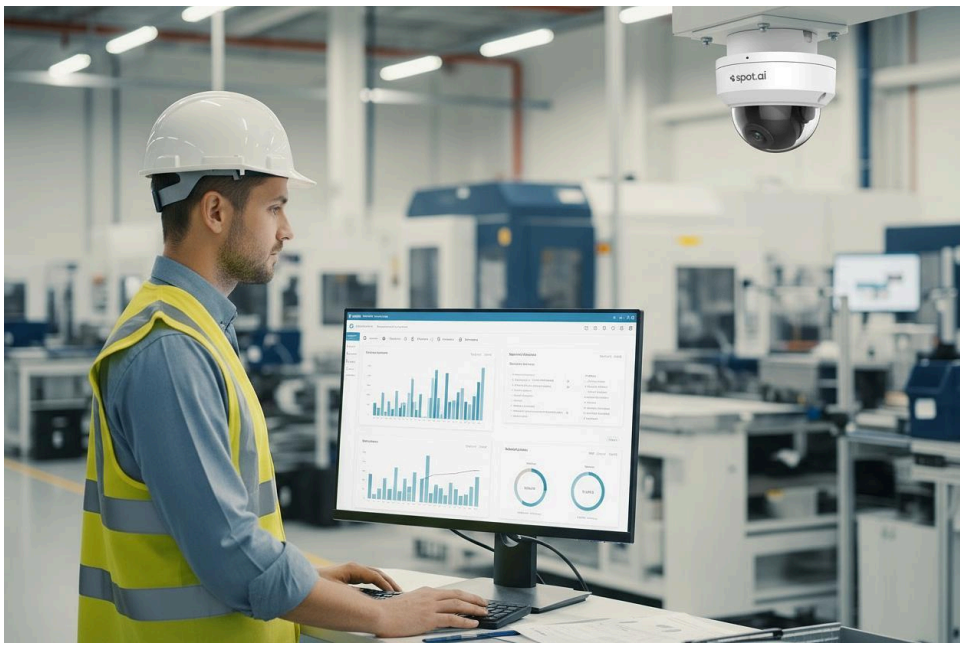
2. Autonomous systems and robotics researchers

Study how perception systems respond to sensor dropout, occlusion and conflicting measurements. This is a relevant future application, although it requires different datasets and validation.



3. Educational AI developers

Build classroom or learning assistants that cope with noisy speech, incomplete lecture recordings and unreliable text extraction.



4. Industrial monitoring teams

Investigate reliable decisions when different sensors provide degraded or inconsistent evidence.

My recommendation: use a multimodal sentiment or emotion-recognition benchmark as the initial controlled research environment. Build the demo around an assistive or general evidence-monitoring scenario only if it can be validated without making unsupported safety claims.

This separation matters: the dataset used for controlled experiments does not have to be the final application.

8. What data do we need, and where will it come from?

The initial plan is to use public multimodal datasets with aligned text, audio and visual information.

Dataset	What it provides	Proposed use
CMU-MOSI	2,199 opinion-video clips with sentiment labels and audio/visual features	Small, controlled initial experiments
CMU-MOSEI	A larger collection of online videos with sentiment and emotion annotations	Main robustness experiments and scaling
MELD	Multimodal dialogue with emotion and sentiment annotations	Dialogue-oriented validation and transfer testing
MMMC	A benchmark focused on modality conflict in multimodal models	Conflict-focused experiments, subject to task and format compatibility

The CMU MultiComp Lab provides official resources for MOSI and MOSEI, while its multimodal SDK supports dataset access and processing.

What about the failure labels?

Most existing datasets will not provide the exact labels we need for every kind of failure. Therefore, we can create a controlled degradation dataset from the original samples.

For each sample, record:

- Original sample ID and split
- Which modality was altered
- Type of degradation
- Severity
- Whether several failures were combined
- Ground-truth task label
- Ground-truth injected failure
- Action taken by the model

- Whether the action improved the outcome

For example:

Example experiment record

Image

30% occlusion

Audio

Missing

Text

Unmodified

Failure setting

Two-way compound

Task label

Original dataset label

What if data is unavailable or invalid?

We need explicit fallback rules:

- If a modality is absent in the original dataset, do not pretend that it can be evaluated as if it were present and clean.
- If labels for a failure type are unavailable, use controlled injection to establish known failure ground truth.
- If a dataset does not support a proposed modality or action, remove that experiment or use a compatible dataset.
- If a synthetic result does not transfer to naturally occurring degradation, limit the claim to the synthetic setting.

A synthetic dataset is useful for controlled diagnosis, but it is not sufficient evidence of real-world reliability.

9. How will we make the approach different?

This is the central research comparison.

Conventional reliability-weighting approach

Multimodal inputs

Estimate reliability

Calculate continuous weights

Fuse representations

Prediction

The main question: how much should each modality contribute?

MULTI-SHIELD research hypothesis

Multimodal inputs

Estimate evidence quality and failure state

Select an action based on the state

Reweight · Suppress · Repair · Defer

Prediction or abstention

The research question: does knowing the type of failure improve the choice of intervention?

The important distinction

We are not claiming that existing methods literally cannot suppress or abstain. Some already support related capabilities.

Our question is narrower:

Does explicitly modelling failure states and conditioning intervention selection on those states outperform a strong, fairly tuned continuous-weighting baseline under compound degradation?

That is the claim we can test.

Why might our approach be better?

There are three possible advantages to investigate:

1. More appropriate intervention: the system can learn that different conditions may favour different actions.
2. Better handling of interacting failures: the policy can consider multiple degraded modalities jointly rather than treating each failure in isolation.
3. More responsible decisions: the system can defer when the evidence is inadequate, rather than forcing a prediction.

But there are also disadvantages. A policy router adds complexity, may misclassify failures, and can select an incorrect repair or suppress useful evidence. We must measure those costs rather than assume the architecture is superior.

10. What is the greatest contribution we could make?

The most important part is not the interface, the number of modules or the four action names.

It is the experimental demonstration that failure-state-conditioned intervention selection provides a measurable advantage over existing alternatives.

I would organise the contribution into three levels.

Primary contribution

A failure-aware decision policy

Test whether a multi-class failure-state representation improves intervention selection under interacting degradation.

Supporting contribution

A reproducible compound-degradation evaluation framework

Evaluate missingness, noise, corruption, conflict and combinations using controlled severity, repeatable splits and documented metadata.

Practical contribution

An explainable reliability dashboard

Show the detected condition, chosen action, confidence, latency and the outcome relative to a baseline.

A further potential contribution is the distinction between valid semantic divergence and genuine evidence failure. It becomes a research contribution only if we can define it rigorously, label it reliably and demonstrate that the distinction improves decisions.

The greatest possible result

Suppose we show that, on unseen compound-degradation conditions:

- MULTI-SHIELD achieves lower selective risk at comparable coverage.
- It maintains competitive Macro-F1 on clean data.
- It chooses interventions more effectively than a tuned scalar baseline.
- The improvement transfers to another dataset.
- Its additional computation is justified by the benefit.

That would constitute meaningful evidence for the approach.

If the policy does not outperform the baseline, we should report that honestly and investigate when the simpler approach is preferable.

11. The experimental plan: how we prove or disprove the hypothesis

We should implement the simplest system first and increase complexity only when evidence justifies it.

1. Build a clean-data baseline
   Train and evaluate a standard multimodal fusion model. Establish Macro-F1, accuracy, calibration, runtime and per-modality performance.
2. Build a strong scalar-weighting baseline
   Implement and tune reliability-based weighting. Include relevant published methods where feasible, with faithful implementations or clearly documented adaptations.
3. Create a controlled degradation generator
   Inject missing modalities, noise, visual occlusion, feature corruption and controlled conflict. Test individual failures first.
4. Combine failure types
   Test two-way and three-way combinations across multiple severities. Keep training and evaluation conditions separate to prevent leakage.
5. Implement and train the policy
   Estimate failure states and choose between reweighting, suppression, repair and deferral. Use training-only data to learn action selection; never supply the known test-time corruption label to the model.
6. Compare outcomes
   Run the same test samples through all methods. Measure accuracy, risk, coverage, calibration, diagnosis quality, action-selection quality and computation.

7. Test generalisation
   Evaluate on another compatible dataset and, where possible, naturally occurring degradation. Clearly distinguish synthetic robustness from real-world validation.
8. Run ablations and statistical tests
   Remove individual diagnostic signals and actions, compare across random seeds and report confidence intervals and paired statistical comparisons.

The central experiment

The main comparison is:

$$Scalarreliabilityweightingvs.Failure - state - conditionedpolicy$$

We should evaluate it separately under:

Test condition	Purpose
Clean inputs	Ensure the policy does not harm normal performance
Single failure	Understand individual failure responses
Two-way compound failure	Test interacting failure mechanisms
Three-way compound failure	Stress-test the core hypothesis
Valid semantic divergence	Test whether useful disagreement is preserved
Unseen degradation severity	Test generalisation beyond memorised conditions
Cross-dataset evaluation	Test whether gains transfer beyond one dataset

This follows the established need to benchmark multimodal robustness under missing and noisy modalities, while extending the experimental question toward action selection under interacting failures.

12. How do we compare MULTI-SHIELD against existing approaches?

We need to distinguish what the method predicts from how safely it makes that prediction.

Metric	What it tells us
Macro-F1	Is the final classification good across classes?
Accuracy	How often is the prediction correct?
AURC	How does selective prediction risk change as the system covers more samples? Lower is generally better under the same protocol.
Risk at fixed coverage	Is MULTI-SHIELD less error-prone when both methods answer the same proportion of cases?
Expected Calibration Error (ECE)	Do confidence estimates align with observed correctness, subject to the limitations of ECE?
Brier score	How good are the probabilistic predictions?
Failure-state macro-F1	Can the system identify failure types?
Action-selection performance	Does it choose useful interventions?

Clean-to-corrupted performance drop	How sensitive is the model to degradation?
Latency and memory	What does the policy cost computationally?

AURC should not be used alone. Selective risk must be compared alongside coverage, and abstention must not be allowed to make a method appear excellent simply by refusing to answer nearly everything.

The comparison rules

- Use identical data splits and comparable model backbones.
- Tune baselines using validation data, not the test set.
- Include strong weighting baselines, not just fixed fusion.
- Report mean performance, variability and confidence intervals across repeated runs.
- Account for multiple statistical comparisons where necessary.
- Report negative results, including conditions in which the policy loses.

What would count as success?

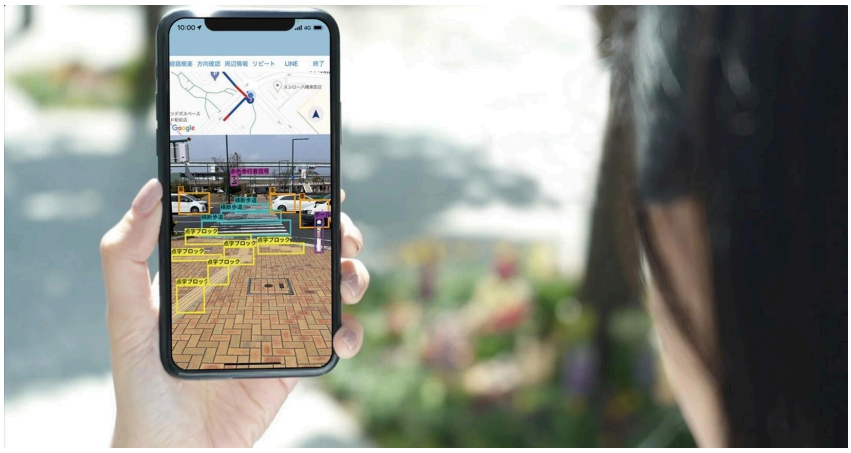
The target is not an arbitrary promise of 5% or 10% improvement.

Success means demonstrating a reproducible, statistically supported and practically meaningful advantage on the predefined primary metrics, without an unacceptable clean-data or computational penalty.

The exact minimum effect size and decision thresholds should be justified and preregistered before the final test runs.

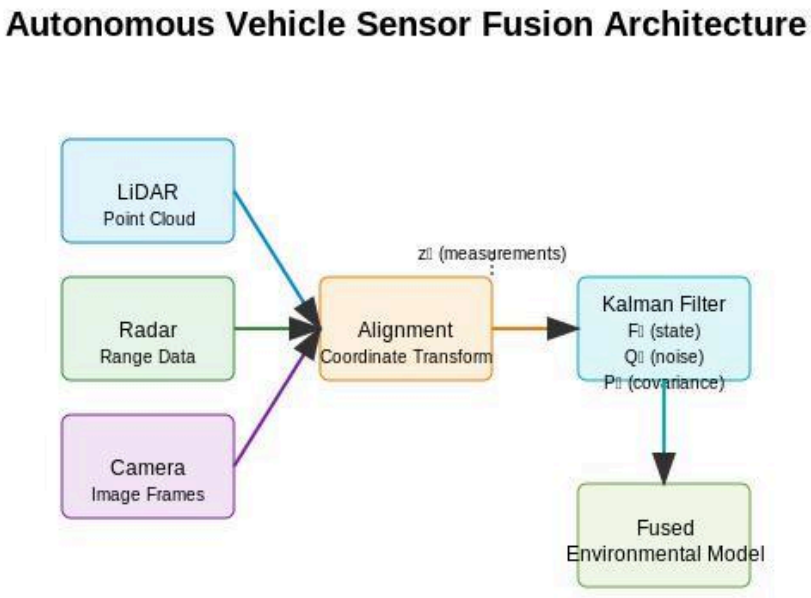
13. What real-world impact could it create?

The impact comes from reducing the harm caused by unreliable evidence—not merely increasing benchmark scores.



Safer assistive interactions

A system could recognise when its visual or speech evidence is inadequate and avoid presenting an unsupported answer. Actual safety benefits would require user studies and application-specific validation.



More resilient perception systems

The research could help developers evaluate how perception systems behave under sensor failures and identify when the available evidence is insufficient.



Better failure diagnosis

Engineers could see which modality was flagged, what intervention was selected and how the decision changed. This makes failures easier to investigate.



Reusable research infrastructure

A documented benchmark and reproducible evaluation pipeline could help other researchers compare reliability strategies under controlled compound degradation.

How can we make the impact stronger?

Prioritise these improvements in order:

1. Demonstrate a genuine performance or risk advantage. A reliable result is more valuable than a complicated architecture.
2. Release reproducible experiments. Provide corruption generators, configurations, evaluation scripts and model checkpoints where licensing permits.
3. Test generalisation. Show whether gains persist across datasets, failure severities and unseen combinations.
4. Make the decisions explainable. Display detected failures, intervention choices and uncertainty without claiming that the explanation is a perfect causal account.
5. Validate one application. Work with a real user group or deployment partner before claiming practical safety or accessibility benefits.

SDG alignment: SDG 9 (Industry, Innovation and Infrastructure) is the strongest general alignment. SDG 10 (Reduced Inequalities) becomes relevant if we validate an assistive-technology application. Neither should be presented as an impact already achieved.

14. What are the risks, and how do we handle them?

Risk	What could go wrong?	Mitigation
Novelty overlap	A published method already implements similar action routing	Maintain an updated literature matrix and compare the closest methods directly
Incorrect diagnosis	Valid evidence is classified as corruption	Evaluate diagnosis separately and include semantic-divergence tests
Repair makes things worse	Reconstruction introduces false information	Reassess the repaired representation and compare against suppression
Policy offers no advantage	Scalar weighting performs equally well	Report the result; simplify the model if routing is not justified

Synthetic-only improvement	Gains disappear on natural data	Add cross-dataset and natural-degradation validation; limit claims if transfer fails
Excessive computation	Multiple diagnostics and repair make inference too slow	Measure latency and memory, then prune expensive components
Dataset mismatch	Existing datasets do not represent the intended application	Separate the research benchmark from the application demonstration
Overstated safety	Better benchmark metrics are treated as proof of deployment safety	Make application-specific claims only after appropriate validation

Our kill condition

If a strong scalar baseline performs equally well, the central hypothesis is not supported.

If a close prior paper already solves the same problem under comparable conditions, we must narrow or change the research question.

If gains occur only under our own synthetic corruption patterns, we must not generalise them to real-world deployments.

These are not reasons to abandon research prematurely. They are rules that prevent us from defending a weak conclusion.

15. What is the scope of the project?

In scope

- Image, audio and text as the initial modalities.
- Missing, noisy, corrupted and conflicting inputs.
- Explicit failure-state estimation.
- Reweighting, suppression, repair and deferral.
- Controlled single and compound degradation.
- Comparisons with strong baseline methods.
- Calibration, selective risk, robustness and computational evaluation.
- An explainable research dashboard.

Out of scope for the first version

- Training a new foundation model from scratch.
- Solving every possible modality failure.
- Guaranteeing that every failure is diagnosed correctly.
- Claiming that the system is safe for autonomous driving, medical decisions or other high-stakes deployment without dedicated validation.
- Proving that all existing multimodal approaches are inferior.

Keeping this scope controlled is essential. The project already has enough research depth without attempting to solve every form of trustworthy AI.

16. What research papers should we use?

The references below combine papers already central to the project with independently located sources relevant to its current direction. The distinction between verified bibliographic information and a claimed research gap is important.

Core prior-art references

1. PRIME — Adaptive Modality Reliability Diagnosis and Restoration for Robust Multimodal Intent Recognition (2026)

Suraj Kumar, Mohnish Raj, Soumi Chattopadhayay, Chandranath Adak and Ayan Dutta. arXiv preprint.

Why it matters: explicit reliability diagnosis, prototype-conditioned restoration, reassessment and reliability-weighted fusion. This is one of the closest approaches to compare.

Read PRIME on arXiv

2. Dynamic Modality Scheduling for Multimodal Large Models via Confidence, Uncertainty, and Semantic Consistency (2025)

Hiroshi Tanaka, Anika Rao, Hana Satou, Michael Johnson and Sofia García. arXiv preprint.

Why it matters: adaptive per-sample modality weighting using predictive confidence, uncertainty and semantic consistency.

Read DMS on arXiv

3. Conformal Fusion Under Missing Modalities (2026)

Alireza Moayedikia. arXiv preprint.

Why it matters: missing-modality robustness, evidential fusion and conformal calibration. It is an important comparison for uncertainty and selective decision-making.

Read MCCF on arXiv

4. Multimodal Fusion on Low-Quality Data: A Comprehensive Survey (2026)

Qingyang Zhang and co-authors. Information Fusion.

Why it matters: provides a broader taxonomy of noisy, incomplete, imbalanced and quality-varying multimodal data. Useful for literature-review coverage and gap verification.

Read the survey

5. Deep Multimodal Learning with Missing Modality: A Survey (2024)

Renjie Wu, Hu Wang, Hsiang-Ting Chen and Gustavo Carneiro. arXiv:2409.07825.

Why it matters: establishes the breadth of missing-modality research and helps prevent us from treating missing-input handling as a new contribution.

Read the missing-modality survey

6. Resilient Sensor Fusion under Adverse Sensor Failures via Multi-Modal Expert Fusion (CVPR 2025)

Konyul Park, Yecheol Kim, Daehun Kim and Jun Won Choi.

Why it matters: directly studies sensor-failure robustness in autonomous-driving perception, including LiDAR failures and camera occlusion.

Read the CVPR paper

7. Benchmarking Multi-modal Semantic Segmentation under Sensor Failures: Missing and Noisy Modality Robustness (CVPR Workshops 2025)

Why it matters: provides a useful precedent for controlled missing/noisy modality evaluation and robustness metrics.

Read the benchmark paper

Dataset and tooling references

- CMU-MOSI official resource
- CMU MultiBench datasets
- CMU Multimodal SDK

These provide starting points for accessing and processing benchmark data.

Important: Before citing any reference in a submitted paper, verify its author list, title, publication status, methodology and actual results from the original source. An arXiv preprint is not automatically peer-reviewed. Also, the Phase 0 candidates MA-AFS, MMMC and SIEVES still need to be checked against their original publications before we make detailed claims about their limitations.

17. How can we turn this into two research papers?

Two papers are possible, but they should answer distinct questions. They must not simply divide one set of experiments into two publications.

Paper 1 · Methodology

Failure-State-Conditioned Decision Policies for Robust Multimodal Learning

Research question: Does explicit failure-state reasoning improve intervention selection and prediction reliability under compound degradation?

Core experiments

- Strong scalar-weighting versus policy-based methods
- Single versus compound degradation
- Reweighting, suppression, repair and deferral
- Calibration, selective risk and action-selection quality
- Ablation and statistical significance

Expected contribution: A rigorously evaluated policy architecture and evidence about when it helps.

Paper 2 · Evaluation / Generalisation

Evaluating Multimodal Failure Diagnosis under Compound Degradation and Semantic Divergence

Research question: How reliably can multimodal systems distinguish genuine input failure from valid semantic divergence, and how does that diagnosis affect downstream decisions?

Potential experiments

- Failure-state classification and confusion matrices
- Valid divergence versus injected corruption
- Cross-dataset generalisation
- Unseen degradation combinations
- Benchmark reproducibility and diagnosis ablations

Expected contribution: A well-defined evaluation protocol and new empirical findings on failure diagnosis.

Paper 2 is conditional. If the diagnosis experiments do not yield an independent methodological or empirical contribution, it should remain part of Paper 1 rather than being forced into a separate publication.

Neither paper is guaranteed to win an award or be accepted by IEEE. The quality of the evidence, novelty relative to prior art, experimental discipline and fit with the venue will determine its prospects.

18. How well does MULTI-SHIELD compare with existing approaches?

The honest answer is that we do not know yet. Here is what we should expect to establish.

Comparison	What we need to demonstrate	If successful
Fixed fusion	Better degradation resilience	Shows the value of adapting to input quality
DMS / other scalar weighting	Better risk or robustness under compound failure	Supports the central policy hypothesis

PRIME	Better decisions in conditions where different interventions are needed	Establishes a defensible distinction from reliability diagnosis and restoration
MCCF	Complementary value beyond missingness calibration	Establishes whether action routing adds something beyond calibrated uncertainty
Conflict-focused methods	Better handling of valid divergence versus genuine failure	Supports the semantic-diagnosis component
Selective prediction methods	Better risk-coverage trade-off	Supports the deferral policy

These are comparison targets, not claims that MULTI-SHIELD currently outperforms them.

Three possible outcomes

Best case

Policy routing produces statistically significant improvements in selective risk and compound robustness, retains clean performance and generalises to another dataset. This supports a strong research contribution.

Mixed result

The policy helps only for particular failure combinations or under particular cost constraints. Narrow the contribution to those conditions and explain when routing is useful.

Negative result

A strong scalar baseline performs equally well or better. Simplify the model, investigate why, and reconsider the research question if necessary.

A negative result is not automatically a failed project. An unsupported novelty claim is the greater risk.

19. How to make the final demonstration impressive

The dashboard should make the scientific result understandable in seconds.

MULTI-SHIELD — Reliability Monitor

Illustrative interface concept. These are example values, not experimental results.

Input health

Image

Corrupted

Audio

Missing

Text

Usable

Selected action

Defer — insufficient evidence

Illustrative decision: image evidence is damaged, audio is unavailable and the remaining evidence does not justify a confident prediction.

Comparison panel

Fixed fusion

To be measured

Scalar weighting

To be measured

MULTI-SHIELD

To be measured

The most valuable feature is the ability to replay the same sample under different failure conditions and compare the decisions of different methods.

The interface should show:

- Original and degraded inputs
- Failure state and estimated uncertainty
- Selected intervention and reason
- Final prediction or deferral
- Baseline comparison
- Risk, coverage and latency
- Whether the intervention actually improved the result

This transforms an abstract research contribution into something your guide, teammates, reviewers and interviewers can inspect.

20. The execution roadmap

Given the scope of your fourth-year major project, the sensible sequence is:

1. Research audit
   Verify the closest papers, build the final literature matrix and confirm the specific research gap.
2. Dataset and baseline
   Obtain a compatible dataset, establish reproducible splits and implement standard fusion.
3. Strong scalar baseline
   Implement and tune reliability-based weighting before developing the proposed policy.
4. Failure generator
   Build controlled missingness, noise, corruption, conflict and compound-failure conditions.
5. Policy prototype
   Implement diagnosis and the four candidate actions. Validate each action independently before training the router.
6. Main experiments
   Run paired comparisons, repeated seeds, ablations, calibration and selective-risk evaluation.
7. Generalisation and demonstration
   Test another dataset if feasible, document limitations and build the dashboard around actual results.
8. Publications and presentation
   Write the primary paper first. Develop a second paper only if the diagnosis or generalisation study supports an independent contribution.

Final verdict

MULTI-SHIELD has a credible research direction, a clear experimental hypothesis, a demonstrable interface and a practical problem worth studying. But the broad field of multimodal reliability is active, and recent methods already cover several of the components we originally intended to claim as novel.

The project should therefore succeed or fail on one central question:

> Can explicitly identifying the type of multimodal failure help an AI system select a better intervention than a strong continuous-weighting approach, especially when multiple failures interact?

If we can demonstrate that convincingly, the project becomes more than a combination of existing techniques. If we cannot, we must narrow the claim or redesign the approach.

That is the standard we should hold it to.